

Zeeshan Ali

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EDUCATION

University of Georgia, College of Engineering – Athens, GA

May 2028

Bachelor of Science in Computer Engineering

Deans's List, GPA: 3.8

Relevant Coursework: Logic Design, Software Development, Computing and Programming, Data Science, Circuits

PROFESSIONAL EXPERIENCE

Ajanta Electrical Industries – *Electrical & Controls Development Intern* – India

June 2025 – August 2025

- Presented an AI-powered Smart Electrical Panel prototype integrating 5 IoT sensors, thermal cameras, and predictive maintenance models for industrial automation, monitoring switchgear, busbars, breakers, and control panel health.
- Observed industrial electrical manufacturing at a 25-year ISO 9001–certified facility serving BHEL and Hindustan Petroleum, focusing on industrial automation systems, quality assurance, and compliance in control panel production.
- Created a detailed Assembly Flow Diagram mapping the end-to-end manufacturing process of industrial control panels, covering component selection, busbar fabrication, and wiring layouts. Presenting to a panel of 10 Executive Members.
- Hands-on experience with LT control panels and MCCs, covering design, wiring, commissioning, and key components used.

Qutbi Mosque – *Civil/Electrical Engineering Intern* – India

June 2024 – August 2024

- Studied electrical system design and power distribution for a large-scale construction project, circuit routing, lighting layouts, and electrical safety standards during an intensive 6 week hands-on engineering internship.
- Reviewed and interpreted multi-disciplinary construction drawings, emphasizing electrical layouts and their coordination with architectural, structural, HVAC, and plumbing plans to ensure constructability and integration across building systems.
- Shadowed electrical and civil site engineers to observe field execution of electrical infrastructure, including conduit routing

PROJECT EXPERIENCE

AI Electronic Waste Sorter Institute of Electrical and Electronics Engineers – Athens, GA

September 2025 – Present

- Developed a Raspberry Pi–based industrial automation prototype integrating cameras, sensors, and GPIO control to detect e-waste in real time, validating electrical constraints, power distribution, and reliable hardware–software interaction.
- Tested, calibrated, and automated DC motors and servo actuators using PWM and GPIO control, coordinating vision-based detection with precise motion control, sequencing, and small-scale industrial automation systems.
- Designed an automated e-waste sorting system from concept to deployment, planning electrical architecture, motor control, sensing, actuation, and control logic to deliver an autonomous, scalable industrial-style material handling solution.

Visual Detection Autonomous Goalie Showcase Project – Milton, GA

January 2023 – May 2023

- Developed an autonomous robotic control system using C++ , integrating Limelight vision, motor controllers, and drivetrain electronics to detect objects, compute positional offsets, and dynamically regulate motor output for precise motion control.
- Converted a manually driven robot into a fully autonomous system by designing closed-loop control architecture linking vision sensors, gyro feedback, control algorithms, and electrical motor actuation for accurate, repeatable robotic movement.
- Designed, tested, and demonstrated a complete autonomous robotic platform, explaining electrical subsystems, sensor integration, motor control, and real-time mechanisms to 300+ students and industry sponsors at an engineering showcase.

CAMPUS INVOLVEMENT & ASSOCIATIONS

MicroMouse Competition Institute of Electrical and Electronics Engineers (IEEE) – Athens, GA

August 2025 – Present

- Designed, simulated, and validated IR sensor circuits using MultiSim and breadboard testing, analyzing voltage variation with distance, then converted working designs into Altium/Fusion schematics for PCB-ready embedded systems.
- Researched and evaluated DC motor types and H-bridge drivers, comparing torque, efficiency, and control tradeoffs to select optimal motors for robotic motion, while applying bidirectional motor control principles in embedded hardware design.
- Studied accelerometers, gyroscopes, and integrated IMUs, reviewing datasheets to understand sensor fusion, schematic requirements, decoupling, and PCB layout considerations for accurate motion sensing in embedded systems.
- Analyzed buck converters and power regulation, optimizing switching efficiency for stable embedded power systems.

TECHNICAL PROFICIENCIES

Hardware: Altium, Autodesk Fusion, Circuit Design, SolidWorks, Fabrication, Power Tools

Software: Python, C++ , Java, MATLAB, Multisim, Microcontrollers, Raspberry Pi, Arduino, GitHub, Microsoft 365 Applications